MSBA 511: Data Mining for Business Analytics

Conceptual Homework 5: Naive Bayes

Objective

Be able to:

- Apply the steps of the Naive Bayes classification algorithm to calculate probabilities.

Overview

A bank is looking to improve its conversion rate for personal loan acceptance in its next marketing campaign. Among the 5000 customers, only 480 (= 9.6%) accepted the personal loan that was offered to them in the prior campaign. In this exercise, we focus on two predictors: whether the customer also has an open CD Account (CD) and Securities Account (SA), and the outcome Personal Loan (PL).

Tasks

1. Calculate the exact probability for: P(PL = 1 | CD = 1, SA = 1
2. Calculate the probability for personal loan acceptance: P(PL = 1)
3. Calculate the probability for personal loan declined: P(PL = 0)
4. Calculate the conditional probability for CD Account given loan acceptance: P(CD = 1 | PL = 1)
5. Calculate the conditional probability for Securities Account given loan acceptance: P(SA = 1 | PL = 1)
6. Calculate the conditional probability for CD Account given loan declined: P(CD = 1 | PL = 0)
7. Calculate the conditional probability for Securities Account given loan declined: P(SA = 1 | PL = 0)
8. Calculate the Naive Bayes Probability: P(PL = 1 | CD = 1, SA = 1)
9. Explain why we may have observed a difference in the exact probability for Personal Loan Acceptance if the customer has a CD Account and Securities Account compared to the Naive Bayes probability.

Submission Requirements

This assignment is completed as a Quiz on Canvas. The Quiz allows 2 attempts and the highest score will be used.